

		move in the same direction as the Earth or the satellite would be moving in an orbit which changes latitudes as it circles around the Earth.
12	B	For (monatomic) ideal gases, $U = \frac{3}{2} NkT$ N: no. of molecules For 1 molecule, average translational KE = $\frac{1}{2} m \langle c^2 \rangle$ m: mass of 1 molecule Hence for ideal gas, Since $U = KE$ only, $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$ $\sqrt{\langle c^2 \rangle} \propto \sqrt{T} \propto \sqrt{PV}$ $\frac{\sqrt{\langle c^2 \rangle \text{ at } T_2}}{\sqrt{\langle c^2 \rangle \text{ at } T_1}} = \frac{\sqrt{PV} \text{ at } T_2}{\sqrt{PV} \text{ at } T_1} = \frac{\sqrt{4}}{\sqrt{1}} = 2$
13	D	Once gas is released, it expands and achieves atmospheric pressure eventually. Using ideal gas equation, pressure is same before and after, but new temperature is 80°C